Formaldehyde Sensor

（SC12-CH_2O）

datasheets

深圳市深晨科技有限公司

Shenzhen Shenchen Technology Co., LTD

Electrochemical Formaldehyde Sensor SC12-CH_2O

Summarize

SC12-CH_2O electrochemical formaldehyde sensor is a general-type, miniaturized module.The detection of CH2O in air has good selectivity and stability.Built-in temperature sensor, temperature compensation; both digital output and analog voltage output, easy to use.SC01-CH2O is a mature electrochemical detection technology with sophisticated circuit design Closely combined, designed and manufactured by a universal gas module.

Design Features

High Sensitivity,Stability,High Reliability Linearity

UART/ PWM Output

Low Power Consumption for Battery Operation

Main application

Portable instrument, air quality monitoring equipment, air purifier, fresh air ventilation system,

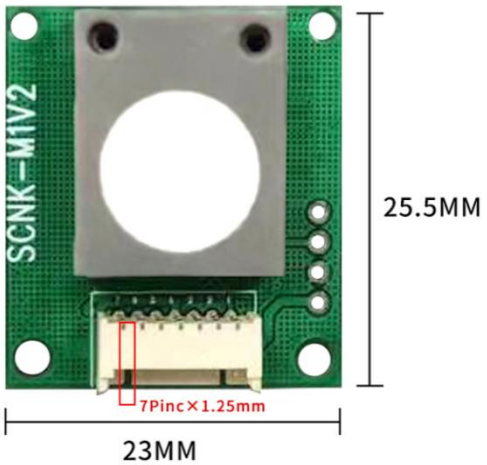
air conditioning, smart home equipment and other places.

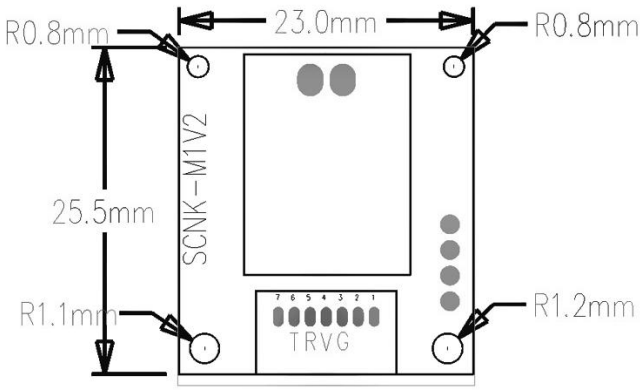
Specifications

Table1

Detection Gas	Formaldehyde
Output Signal	UART （3.3V ）/ PWM
Operating voltage	4.2V～5.5V
Preheating time	≤3 min
Typical Response Time (t90)	≤60 S
Recovery time	≤60 S
Detection Range	0～2000 PPB
Resolution ratio	≤ 1PPB
Operating Temperature	-20℃～50℃
Operating Humidity	15%RH-90％RH（No condensation）
Recommended Storage Temp.	0～25℃
Expected Life Time	2 Year（in the air 0℃～35℃）
Size	L 23mm × W 25.5mm × H 6.5mm

Size（unit：mm）





Pin Definition

Table 2

PIN	Definition
Pin1	PWM
Pin2	NC
Pin3	GND
Pin4	Vin（ 4.2V～5.5V）
Pin5	UART（RXD） 0～3V data-in
Pin6	UART（TXD） 0～3V data-out
Pin7	NC

Picture 2： Module pin

Communications Interface

1. General settings

Table3

Baud rate	9600
data bit	8 bits
stop bit	1 bits
check bit	NO

2. Communication command

Communication is divided into active upload type and Q & A typeFactory default active upload,concentration values were sent at 1S intervals.

If the user uses the Q & A mode，change to using active upload mode,send the following command-line format:

Table 4

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
start bit	Reserved	Switch Command	Take the initiative Upload	Reserved	Reserved	Reserved	Reserved	Check data
0xFF	0x01	0x78	0x40	0x00	0x00	0x00	0x00	0x47

Proactively uploaded data display format：

Table 5

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
start bit	Gas name (O3)	Unit (ppb)	Decimal locus NC	Gas concentration High	Gas concentration Low	Full range High	Full range Low	Check data
0x FF	0x17	0x04	0x00	0x00	0x25	0x13	0x88	0x25

Explanatory note: Gas concentration value(PPB)=(Byte4 *256+Byte5).
When converted to PPM： PPM= PPB/1000.
1PPM CH_2O is equivalent to 1.34mg/m^3. 1mg/m^3 CH_2O is equivalent to 0.746PPM.

When using the Q & A mode,turn off the actively uploaded data by sending the following command format,and send a command to read the concentration.The command-line format for closing the active upload is as follows：

Table 6

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
start bit	Reserved	Change order	Q & A	Reserved	Reserved	Reserved	Reserved	Check data
0xFF	0x01	0x78	0x41	0x00	0x00	0x00	0x00	0x46

Q & A mode ,the command format for the read concentration is as follows：

Table 7

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
start bit	Reserved	order	Reserved	Reserved	Reserved	Reserved	Reserved	Check data
0xFF	0x01	0x86	0x00	0x00	0x00	0x00	0x00	0x79

The sensor concentration values are returned in the following format:：

Table 8

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
start bit	order	Gas concentration High(ug/m3)	Gas concentration Low(ug/m3)	Reserved	Reserved	Gas concentration High(ppb)	Gas concentration Low(ppb)	Check data
0xFF	0x86	0x00	0x2A	0x00	0x00	0x00	0x20	0x30

Gas concentration value=Gas concentration High*256+Gas concentration Low

3. Checksum calculate

check sum = （Refuse（Byte1+Byte2+……+Byte7））+ 1

Reference routines are as follows:

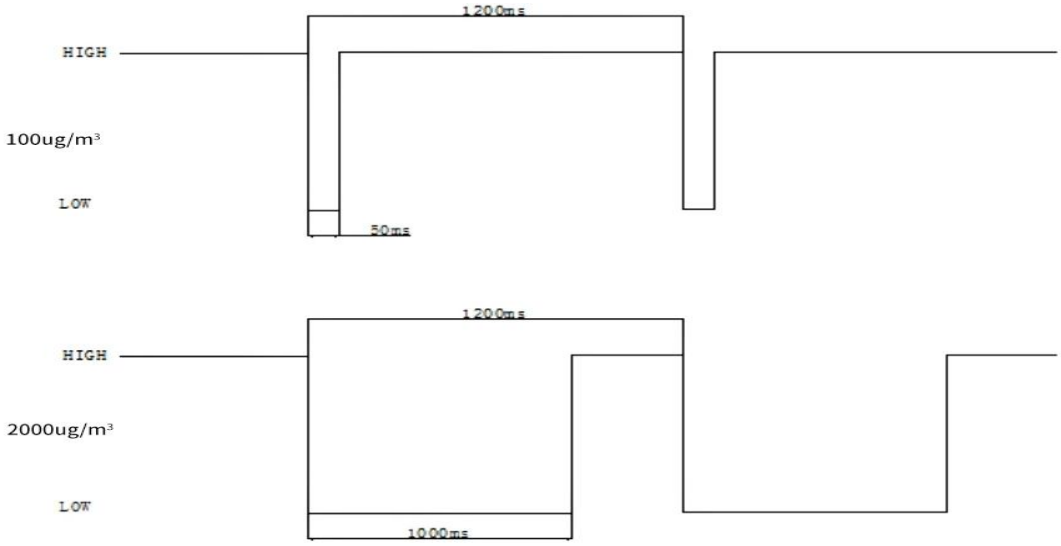
```
/*********************************************************************
* function name : unsigned char FucCheckSum(uchar *i,ucharln)
*  functional description : summation check （Take the reverse + 1 of 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 for
   sending and receiving protocol）
function declaration:（Refuse（Byte1+Byte2+……+Byte7））+ 1
*********************************************************************/
unsigned char FucCheckSum(unsigned char *i,unsigned char ln)
{
    unsigned char
    j,tempq=0; i+=1;
    for(j=0;j<(ln-2);j++)
    {
        tempq+=
        *i; i++;
    }
    tempq=(~tempq)
    +1; return(tempq);
}
```

4. PWM output

The sensor outputs a PWM signal through the PWM port (pin 5), with a PWM period of 1.2s. The formaldehyde concentration value is calculated based on the low-level width.

For example, if the low-level width is 50ms, the corresponding concentration is 100ug/m3.

The concentration output range is from 0ug/m3 to 2000ug/m3



Interference characteristic

The sensor module has excellent interference resistance,There is no response to most gas except the target gas.The response properties of this sensor to several common interference gases are listed in the table below.Data in the table are the typical response of the gas at a given concentration.

Cross Sensitivity of SC01-CH_2O :

Gas	Concentration/PPM	HCHO Equal concentration/PPM
HCHO	5	5
benzene	10	0.1
methylbenzene	10	0.46
acetic acid	200	0.52
alcohol	100	40.6
hepatic gas	50	3
CO	200	0.64

Attention

■ Upper tin welding shall be completed quickly during installation pin welding to avoid damage to the sensor structure by prolonged high temperature；

■ Aging time is not less than 48 hours before use;

■ Electrolysis leakage can cause damage, do not remove the shell of the sensor；

■ The sensor avoids using organic solvents (including silicone rubber and other adhesives), coatings, agents, oils, and high-concentration gases

■ All electrochemical sensors cannot be completely sealed with a resin material， can not be used in an anaerobic environment， otherwise, it will damage the sensor.

■ All electrochemical sensors cannot be used in environments containing corrosive gases， corrosive gas can damage the sensor;；

■ The gas zero-point determination must be conducted in a clean atmosphere.